Analytic State-Averaged DMRG-CASSCF Gradients and Nonadiabatic Couplings in a PySCF/pyblock2 Workflow

Author list to be finalized

April 27, 2026

Abstract

Analytic nuclear gradients and nonadiabatic coupling vectors are central ingredients for locating conical intersections and for propagating nonadiabatic dynamics with multireference electronic-structure methods. Complete-active-space self-consistent-field (CASSCF) theory provides a robust reference near electronic degeneracies, but full configuration interaction in the active space limits the accessible number of active orbitals. Here we report a PySCF/pyblock2 implementation of analytic state-averaged density matrix renormalization group CASSCF (SA-DMRG-CASSCF) gradients and nonadiabatic couplings, together with a SHARC-compatible interface for requesting energies, dipoles, gradients, and derivative couplings from the same workflow. The implementation is validated against PySCF full-CI active-space references on public benchmark systems. We further quantify the finite-bond-dimension response error by replacing FCI active-space vectors with block2 matrix product states at fixed FCI-optimized orbitals. The resulting convergence curves separate intrinsic DMRG compression error from artifacts associated with reoptimizing orbitals in a truncated CI manifold. An ethylene interface regression test demonstrates that the implemented DMRG-CASSCF backend writes all requested SHARC data blocks.

1 Introduction

Nonadiabatic molecular processes require electronic-structure information beyond adiabatic energies. Surface-hopping algorithms and related mixed quantum–classical methods use gradients and state couplings to propagate nuclear motion through regions where the Born–Oppenheimer separation breaks down.[1, 2, 3] For same-spin internal conversion, analytic derivative coupling vectors are especially useful because they define, together with state gradients, the first-order topology near conical intersections and provide direct inputs to optimization and dynamics workflows.[4]

CASSCF and state-averaged CASSCF (SA-CASSCF) wave functions are widely used for this purpose because they provide a balanced multiconfigurational description of near-degenerate states. The conventional active-space solver, however, is a full configuration interaction (FCI) expansion whose dimension grows combinatorially with active electrons and orbitals. The density matrix renormalization group (DMRG) replaces the FCI vector by a matrix product state (MPS), making compact approximations to much larger active spaces possible when the entanglement structure is favorable.[5, 6, 7]

Analytic response theory for DMRG-CASSCF has developed rapidly. Freitag and co-workers introduced an approximate SA-DMRG-SCF gradient and nonadiabatic coupling formalism based on single-site MPS Lagrange multipliers in a mixed-canonical representation.[8] Iino, Shiozaki, and Yanai later formulated analytic nuclear gradients for SA-DMRG-CASSCF using newly derived coupled-perturbed equations.[9] These developments establish the theoretical foundation for

Table 1: Positioning of the present work relative to closely related methods and software.

Work or software	Scope	Relation to the present implementation
SA-CASSCF derivative couplings[4]	Analytic CASSCF NAC theory and implementations	Provides the FCI active-space response reference used for numerical validation.
SA-DMRG-SCF response[8]	Approximate analytic gradients and NACs with MPS response variables	Provides the theoretical DMRG response framework followed here.
SA-DMRG-CASSCF gradients[9]	Coupled-perturbed equations for analytic SA-DMRG-CASSCF gradients	Establishes an independent modern formulation; the present work emphasizes open PySCF/pyblock2 integration and NAC/interface validation.
PySCF and block2[10, 11, 12]	Open electronic-structure and DMRG infrastructure	Supplies the host CASSCF derivative code and the spin-adapted MPS backend.
SHARC[2, 3]	Nonadiabatic dynamics driver and data interface	Defines the Hamiltonian, dipole, gradient, and derivative-coupling output contract tested in the ethylene regression case.

DMRG-based derivative calculations. The remaining practical need addressed here is an open, reproducible PySCF-based implementation with regression tests, public benchmark data, and an interface usable by SHARC-style nonadiabatic dynamics workflows.

This work makes three contributions. First, we implement two complementary SA-DMRG-CASSCF response paths in a PySCF/pyblock2 environment: a native MPS response class and a PySCF-compatible active-space solver wrapper. Second, we validate gradients and nonadiabatic couplings against FCI active-space references and characterize the finite-bond-dimension response error on public benchmark systems. Third, we expose the validated backend through a SHARC-compatible PySCF interface and demonstrate the complete output path on a public ethylene test case. The contribution is therefore an implementation, validation, and reproducibility paper rather than a new derivation of the underlying SA-DMRG response theory.

2 Theory and Algorithm

2.1 State-Averaged CASSCF Response

For an SA-CASSCF wave function, the orbitals and active-space CI coefficients are optimized for a weighted average of several electronic states. Analytic gradients and nonadiabatic couplings are obtained by differentiating a Lagrangian that enforces stationarity with respect to orbital rotations and active-space wave-function parameters. The resulting coupled-perturbed CASSCF equations determine the orbital-response multipliers that enter the effective one- and two-particle density matrices used in the derivative contractions.[4, 8]

2.2 DMRG Active-Space Representation

In DMRG-CASSCF, each active-space state is represented by an MPS rather than by an explicit FCI vector. A finite bond dimension M controls the rank of the MPS approximation and therefore the accuracy of energies, density matrices, gradients, and derivative couplings. Following the SA-DMRG-SCF response strategy of Freitag et al., the present implementation evaluates the response quantities needed by PySCF's SA-CASSCF derivative machinery while using pyblock2/block2 for the MPS representation and DMRG optimization. [8, 12]

2.3 Bond-Dimension Response Error

To isolate the effect of the MPS compression, we define a fixed-orbital bond-dimension response error. For each benchmark system we first converge a standard SA-CASSCF calculation with an FCI active-space solver. We then hold the converged molecular orbitals fixed, run spin-adapted block2 DMRG in the same active-space Hamiltonian at a selected bond dimension M, convert the resulting MPS roots into the PySCF active-space representation, and recompute analytic gradients and nonadiabatic couplings. The error is measured against the FCI active-space derivative at the same orbitals,

$$\Delta_g(M) = \left\| \nabla_{\mathbf{R}} E_0(M) - \nabla_{\mathbf{R}} E_0^{\mathrm{FCI}} \right\|_2 , \tag{1}$$

$$\Delta_d(M) = \min_{\sigma=\pm 1} \left\| \mathbf{d}_{01}(M) - \sigma \mathbf{d}_{01}^{\mathrm{FCI}} \right\|_2 , \tag{2}$$

where the sign minimization removes the arbitrary global phase of the electronic states. This protocol avoids mixing the MPS compression error with nonstationarity from orbital reoptimization on a deliberately truncated CI manifold.

3 Implementation

The implementation is built on PySCF[10, 11] and pyblock2/block2.[12] Two code paths are provided. The native `CPDMRGCASSCFResponseMPS` path keeps the response operations in MPS form. The `MPSAsFCISolver` path exposes a PySCF-compatible active-space solver interface, allowing the same DMRG roots to be used by PySCF's CASSCF gradient and nonadiabatic-coupling infrastructure. The latter path is the main regression-test target because it permits direct comparison with standard FCI active-space calculations.

The workflow used in the public benchmarks is:

1. converge an SA(2)-CASSCF reference with PySCF and an FCI active-space solver;
2. freeze the converged orbitals and construct the active-space Hamiltonian in the same orbital basis;
3. optimize spin-adapted block2 MPS roots at the selected bond dimension;
4. align the MPS roots with the FCI reference by maximum overlap and global phase;
5. evaluate energies, one- and two-particle density matrices, gradients, and derivative couplings through the PySCF response interface; and
6. compare the resulting derivatives to the FCI active-space values at identical orbitals.

The SHARC-facing interface adds a PySCF template method, `dmrg-casscf`. When requested, the interface constructs a PySCF CASSCF object, installs the DMRG-backed active-space solver, evaluates the requested Hamiltonian, dipole, gradient, and derivative-coupling quantities, and writes them in SHARC `QM.out` format. The ethylene example in this work is used as an end-to-end interface regression test.

4 Computational Details

All benchmark calculations used singlet state averaging over two roots with equal weights. The public systems are listed in Table 2. The bond-dimension response-error calculations used FCI-converged SA-CASSCF orbitals as the fixed orbital basis. Spin-adapted SU2 block2 DMRG was then run at each selected bond dimension, and the resulting state vectors were phase-aligned to the FCI reference before evaluating derivative errors. Gradients are reported in mE_h/Bohr and derivative couplings in atomic units.

The FCI reference calculations used restricted Hartree–Fock orbitals converged to 10^{-12}, SA-CASSCF energy convergence of 10^{-10}, and a CASSCF gradient convergence threshold of 10^{-7}. The bond-dimension scan used up to 30 DMRG sweeps and a sweep tolerance of 10^{-12} unless noted otherwise. The ethylene SHARC-interface regression test used an SA(2)-DMRG-CASSCF(2,2)/STO-3G model with $M = 60$, ten DMRG sweeps, a DMRG sweep tolerance of 10^{-8}, CASSCF energy convergence of 10^{-8}, and CASSCF gradient convergence of 10^{-5}. These settings are intentionally small enough for a public regression test and are not used to make a mechanistic chemistry claim.

Table 2: Public validation and benchmark systems.

System	Basis	Active space	Role
H_4 chain	STO-3G	CAS(4,4)	Small-system response validation
H_2O	STO-3G	CAS(4,4)	Small-system response validation
N_2	STO-3G	CAS(6,6)	Medium-active-space response validation
H_2O	6-31G	CAS(6,6)	NAC convergence benchmark
C_2	STO-3G	CAS(8,8)	Larger active-space benchmark
LiF	STO-3G	CAS(4,4)	Nonzero-NAC benchmark
Ethylene	STO-3G	CAS(2,2)	SHARC-interface regression test

5 Results and Discussion

5.1 Validation Against FCI Active-Space References

The implementation is covered by unit and integration tests for DMRG/FCI energies, one- and transition-density matrices, MPS-to-FCI conversion, analytic gradients, analytic derivative couplings, and SHARC output formatting. The current regression suite contains 46 passing tests across 14 test files. These tests verify both direct numerical agreement with FCI active-space references in small active spaces and consistency between the native MPS and PySCF-compatible solver paths.

5.2 Finite-Bond-Dimension Response Error

Figure 1 shows the fixed-orbital bond-dimension response error. The H_4, H_2O/STO-3G, N_2, and H_2O/6-31G curves show systematic decay of the derivative error as M is increased. The CAS(8,8) C_2 system is a more demanding benchmark; it improves substantially with M but also shows the expected sensitivity of excited-state tracking in a compact benchmark. LiF provides a nonzero derivative-coupling test case. Its energy and gradient errors converge rapidly, whereas the derivative coupling remains more gauge-sensitive, so it is interpreted as a diagnostic benchmark rather than the primary convergence example.

Table 3: Validation layers included in the public test and benchmark suite.

Validation target	Public evidence	Quantity checked
DMRG active-space solver	Solver regression tests	Energies, state density matrices, transition density matrices, and analytic NAC pipeline consistency with FCI references.
MPS/FCI mapping	Active-space map tests	Round-trip mapping for CAS(2,2), CAS(4,4), and CAS(8,8) active spaces.
Response linear algebra	Coupled-perturbed response tests	Hermiticity, linearity, response-vector solves, and agreement between MPS and FCI response operations in exactly recoverable cases.
Gradient and NAC derivatives	Derivative regression tests	Analytic gradients and derivative couplings against PySCF FCI active-space references.
SHARC interface	Ethylene interface regression	Presence of `H`, `DM`, `GRAD`, and `NACDR` blocks in `QM.out` with SHARC master error code 0.

Table 4: Representative largest-M fixed-orbital response errors from the public benchmark scan.

System	Active space	M	Δ_g (mE_h/Bohr)	Δ_d
H_4/STO-3G	CAS(4,4)	12	8.99×10^{-11}	2.22×10^{-7}
H_2O/STO-3G	CAS(4,4)	12	1.44×10^{-3}	2.29×10^{-6}
N_2/STO-3G	CAS(6,6)	30	4.58×10^{-1}	3.72×10^{-3}
H_2O/6-31G	CAS(6,6)	30	9.37×10^{-5}	1.48×10^{-4}
C_2/STO-3G	CAS(8,8)	70	1.80	5.70×10^{-2}
LiF/STO-3G	CAS(4,4)	12	1.75×10^{-6}	2.91×10^{-1}

The key observation is that the catastrophic response spikes observed with an earlier SVD-truncated CI-vector proxy disappear when real DMRG roots are evaluated at FCI-stationary orbitals. This supports the interpretation that those spikes were not intrinsic failures of the DMRG response approximation, but artifacts of using a nonstationary truncated-CI orbital manifold.

5.3 SHARC Interface Regression Test

The public ethylene test requests the Hamiltonian, dipole matrices, gradients for both singlet states, and the S_0/S_1 derivative coupling. The calculation completes with SHARC master error code 0 and writes all requested data blocks to `QM.out`. This confirms that the DMRG-backed PySCF object satisfies the data contract expected by the SHARC interface for energies, dipoles, gradients, and derivative couplings.

6 Conclusions

We have implemented analytic SA-DMRG-CASSCF gradients and nonadiabatic couplings in a PySCF/pyblock2 workflow and exposed the backend through a SHARC-compatible PySCF interface. The implementation is validated against FCI active-space references, and public benchmark

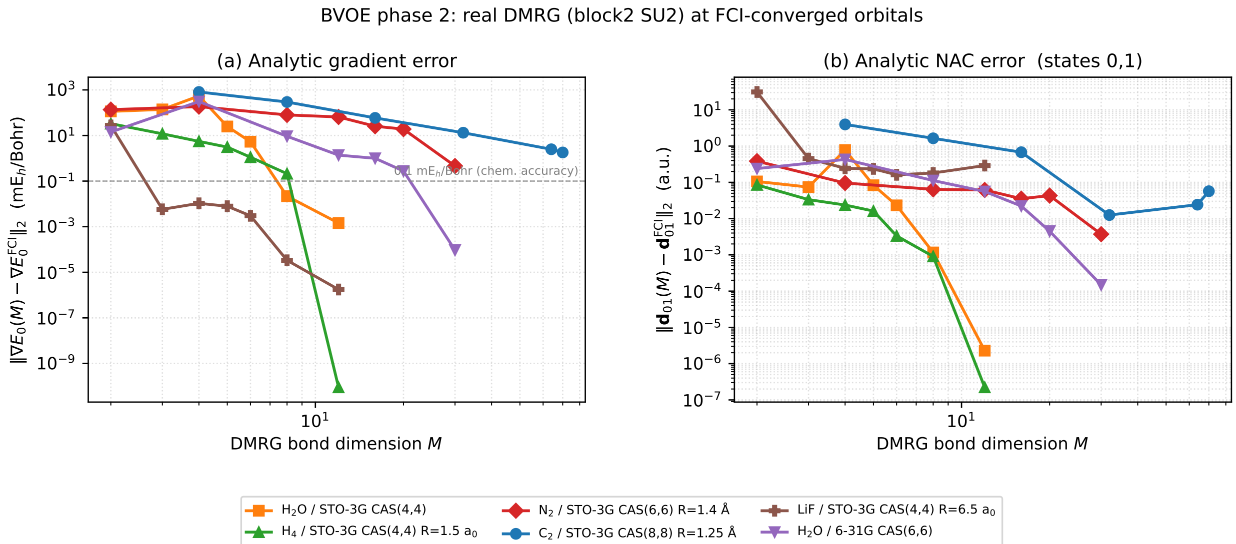

Figure 1: Fixed-orbital bond-dimension response-error convergence for public benchmark systems. Gradient errors are reported in mE_h/Bohr and nonadiabatic coupling errors in atomic units.

data quantify how gradient and derivative-coupling errors decay with DMRG bond dimension at fixed FCI-optimized orbitals. The results provide a reproducible open-source platform for testing DMRG-based multireference response quantities and for connecting them to nonadiabatic dynamics interfaces.

Supporting Information

Supporting Information: benchmark inputs, full convergence tables, regression-test summary, and SHARC ethylene interface example files.

Data and Code Availability

The public code bundle contains the response implementation, tests, benchmark drivers, benchmark summaries, plotting scripts, and the ethylene SHARC interface regression example. A permanent repository URL or DOI should be inserted in this section before journal submission. Application systems and trajectory data outside the public benchmark set are not included.

Acknowledgments

Acknowledgments to be added.

References

[1] John C. Tully. Molecular dynamics with electronic transitions. *Journal of Chemical Physics*, 93:1061–1071, 1990.

[2] Martin Richter, Philipp Marquetand, Jesús González-Vázquez, Ignacio Sola, and Leticia González. SHARC: ab initio molecular dynamics with surface hopping in the adiabatic representation including arbitrary couplings. *Journal of Chemical Theory and Computation*, 7:1253–1258, 2011.

[3] Sebastian Mai, Philipp Marquetand, and Leticia González. Nonadiabatic dynamics: The SHARC approach. *WIREs Computational Molecular Science*, 8:e1370, 2018.

[4] Ignacio Fdez. Galván, Mickaël G. Delcey, Thomas Bondo Pedersen, Francesco Aquilante, and Roland Lindh. Analytical state-average complete-active-space self-consistent field nonadiabatic coupling vectors: Implementation with density-fitted two-electron integrals and application to conical intersections. *Journal of Chemical Theory and Computation*, 12:3636–3653, 2016.

[5] Garnet Kin-Lic Chan and Martin Head-Gordon. Highly correlated calculations with a polynomial cost algorithm: A study of the density matrix renormalization group. *Journal of Chemical Physics*, 116:4462–4476, 2002.

[6] Garnet Kin-Lic Chan and Sandeep Sharma. The density matrix renormalization group in quantum chemistry. *Annual Review of Physical Chemistry*, 62:465–481, 2011.

[7] Sandeep Sharma and Garnet Kin-Lic Chan. Spin-adapted density matrix renormalization group algorithms for quantum chemistry. *Journal of Chemical Physics*, 136:124121, 2012.

[8] Leon Freitag, Yingjin Ma, Alberto Baiardi, Stefan Knecht, and Markus Reiher. Approximate analytical gradients and nonadiabatic couplings for the state-average density matrix renormalization group self-consistent-field method. *Journal of Chemical Theory and Computation*, 15:6724–6737, 2019.

[9] Tsubasa Iino, Toru Shiozaki, and Takeshi Yanai. Algorithm for analytic nuclear energy gradients of state averaged DMRG-CASSCF theory with newly derived coupled-perturbed equations. *Journal of Chemical Physics*, 158:054107, 2023.

[10] Qiming Sun, Timothy C. Berkelbach, Nick S. Blunt, George H. Booth, Sheng Guo, Zhendong Li, Junzi Liu, James D. McClain, Elvira R. Sayfutyarova, Sandeep Sharma, Sebastian Wouters, and Garnet Kin-Lic Chan. PySCF: the python-based simulations of chemistry framework. *WIREs Computational Molecular Science*, 8:e1340, 2018.

[11] Qiming Sun, Xing Zhang, Samragni Banerjee, Peng Bao, Marc Barbry, Nick S. Blunt, Nikolay A. Bogdanov, George H. Booth, Jia Chen, Zhi-Hao Cui, Janus J. Eriksen, Yang Gao, Sheng Guo, Jan Hermann, Matthew R. Hermes, Kevin Koh, Peter Koval, Susi Lehtola, Zhendong Li, Junzi Liu, Narbe Mardirossian, James D. McClain, Mario Motta, Bastien Mussard, Hung Q. Pham, Artem Pulkin, Wirawan Purwanto, Paul J. Robinson, Enrico Ronca, Elvira R. Sayfutyarova, Maximilian Scheurer, Henry F. Schurkus, James E. T. Smith, Chong Sun, Shi-Ning Sun, Shiv Upadhyay, Lucas K. Wagner, Xiao Wang, Alec White, James Daniel Whitfield, Mark J. Williamson, Sebastian Wouters, Jun Yang, Jason M. Yu, Tianyu Zhu, Timothy C. Berkelbach, Sandeep Sharma, Alexander Yu. Sokolov, and Garnet Kin-Lic Chan. Recent developments in the PySCF program package. *Journal of Chemical Physics*, 153:024109, 2020.

[12] Huanchen Zhai, Henrik R. Larsson, Seunghoon Lee, Zhi-Hao Cui, Tianyu Zhu, Chong Sun, Linqing Peng, Ruojing Peng, Ke Liao, Johannes Tölle, Junjie Yang, Shuoxue Li, and Garnet

Kin-Lic Chan. Block2: a comprehensive open source framework to develop and apply state-of-the-art DMRG algorithms in electronic structure and beyond. *Journal of Chemical Physics*, 159:234801, 2023.